CPU Scheduling Criteria

1. **CPU utilization**
   The main objective of any CPU scheduling algorithm is to keep the CPU as busy as possible.
2. **Throughput**
   How many processes the system finishes in a certain time.
3. **Turnaround Time (TAT)**
   Total time from when a task starts until it finishes.
   TAT = Completion Time – Arrival Time
4. **Completion time**
   The time when the process finishes all its execution and is done.
   e.g. If the Word file task ends at 10 seconds, completion time is 10s.
   What is Arrival Time?
   The time when a process enters the system (or is submitted for execution).
   e.g. If you open a Word file at 2 seconds, its arrival time is 2s.
5. **Waiting Time**
   How long a process waits in the ready queue (not running).
   WT = Turnaround Time – Burst Time

   What is **Burst Time? - It** is the time **used by the CPU**
   e.g. A process arrives at 2s, It waits in queue for 3s, Then it runs for 4s (Burst Time = 4s)
   It completes at 2s + 3s + 4s = 9s → Completion Time = 9s

6. **Response Time**
   Response Time = CPU Allocation Time(when the CPU was allocated for the first) - Arrival Time
7. **Priority**
   If the operating system assigns priorities to processes, the scheduling mechanism should favor the higher-priority processes.
8. **Predictability**
   A given process always should run in about the same amount of time under a similar system load.